

Process re-engineering and success of integration projects of Information Technologies

Case study: Process modeling of a cross docking platform of a car manufacturer

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ABSTRACT - The business process reengineering is causing a wave of restructuring and reorganization of companies that largely rely on computer engineering and the potential of information and communications technology. Most reengineering researchers argue that information and communications technology (ICT) have the capacity, through their intrinsic qualities, to change the organization; this is more to computerize the existing but to redefine the work organization and sequencing of tasks and rethink the process. In this research, we present a case study of re-engineering of a platform of cross docking (PCD) using the Supply Chain Operation Reference Model (SCOR) in a car assembly unit. This paper tends to dismantle the relationship between information technology (IT) and process reengineering, also, the success of integration of ICT such as EDI (Electronic Data Interchange), RFID (Radio Frequency Identification) as an approach to achieve the level of performance expected by the logistics chain studied. After presenting the different readings about the process reengineering and the impact of ICT on the performance, we will model the PCD processes by using SCOR and BPMN 2.0.

The purpose of this modelling is to understand at first the functioning of the platforms of cross docking and the interactions between the upstream processes and the downstream of his customers and his suppliers.

In the second stage, this modelling allowed to detect the weaknesses in the processes, further to a lack of traceability in the piloting of the internal flows.

We proposed the integration of the technology RFID for landing in these weaknesses.

We feigned several scenarios before and after the integration of the RFID to validate our hypotheses on the relevance of the solution RFID.

Keywords—*Process re-engineering; SCOR; BPMN; Information technologies; Logistics platform; Modeling; automotive supply chain; RFID; Key performance.*

I. INTRODUCTION

In recent decades, the progress made in the field of information technology (IT) quickly transformed the conditions of realization of transactions worldwide. With their ability to provide timely, accurate and reliable information technology have led to better performance from both the local firm and its partners in the supply chain [1]. The information and communications technology (ICT) are considered, indeed, as the supporting infrastructure not only within the organization but also its upstream; they are also recognized as a factor in the improvement of supply chain management (SCM) [2][3]. Thus, many organizations that focus on the introduction of new information and communications technology and the development of information systems that positively affect their business processes. Yet one of the most important strategic decisions that companies can take is not to determine how to use IT tools to improve their business processes, but what business processes they need to improve. Indeed, a company generally uses hundreds or even thousands of business processes and is usually conducted to reconsider and reconfigure its processes

before implementing a new form of information technology if it wants to achieve realize substantial profits on invested capital and profit performance of the tool or the implanted system[4]. Of course, some projects may require radical reorganization while others are calling for progressive change. In this sense, the new organizational forms combine both a strategic change, merging of socio-technical innovations and structural arrangements, but also a profound change of management and work organization contributing to the company extra flexibility thus opening important fields to increase their competitiveness [5][6][7].

In addition, growing competition created strong pressure for even more functionality, quality and reliability at the same time with requirements to deliver with less cost and fewer timescales [8]. As well, many are the organizations that focus on the introduction of information technologies such as EDI (Electronic Data Interchange), the RFID (Radio Frequency Identification), smart sensors, etc. and on the development of information systems, which affect positively their business processes. One of the most important strategic decisions that companies can take is not to determine how the use of computer tools to improve their business processes, but rather what process they must improve.

We seek, through this study, to demonstrate the relationship between information technology (IT) such as EDI, RFID, GPA (Shared Supply Management), VMI (Vendor Managed Inventory), smart sensors, etc., and process reengineering in a perspective of organizational changes. This paper tends to dismantle the relationship between information technology and business process re-engineering developing how early can generate the second with the transversal management joint. Our problem is thus structured around the following question: How can generate an IT process reengineering serving a transversal managerial? This test will consist of four areas: the first will shed light on the link between IT and process reengineering, develop the second stage of the implementation of reengineering processes, constraints and success factors for successful organizational change planned. The last axis will present a case for modernizing the management of cross docking platform (PCD) through the integration of new technologies with a discussion around the proposed model.

This paper presents a study on the integration of new technology approach, to a logistics platform (PL) or a platform of cross-docking (PCD) under the guidance of a car manufacturer. This integration aims to improve the management and performance of the PCD. This approach cannot succeed without process reengineering able to accurately describe the activities and tasks that these new technologies can make from the current operation.

To do this, we need to understand the current functioning of this PCD, detect weaknesses and then to provide new technologies that can improve these weaknesses. The first part of our paper will be devoted to a review of the literature on process reengineering, and the impact of information technology on performance.

We must not forget that our ultimate goal is to improve the performance of PCD, hence the need to look for tools or models able to provide performance indicators. It is for this reason that we have chosen the SCOR model. A quick review of the literature on this model will be treated in the second part of this paper.

To mitigate the static aspect of the model SCOR, in particular on the return flows, we used the BPMN tool.

The third part will be devoted to the presentation and discussion of the modeling results. Through the last part, we will present our own approach we consider it essential for the success of the integration project of new technologies in the logistics platform studied.

We proposed the integration of the RFID for a better visibility on business process.

We feigned several scenarios to validate our model before and after the integration of the RFID.

II. IMPLEMENTATION OF PROCESS REENGINEERING: STEPS, CONSTRAINTS, AND SUCCESS FACTORS

2.1 *Origins and definitions of concepts*

Hammer and Champy in [9] define reengineering business processes (Business Process Reengineering) as "the fundamental questioning and radical redesign of organizational processes, to achieve dramatic improvements in current performance in cost, services and speed ". Rather than organizing a firm into functional divisions (like production, accounting, marketing, etc.) and to examine the tasks that each function performs, the two authors recommend focusing on the complete processes: acquisition materials, through production, to marketing and distribution. The organization should be rebuilt around a series of processes. The business process reengineering is causing a wave of restructuring and reorganization of companies that largely rely on computer engineering and the potential of information and communications technology. Historically and since the early 90s, the business process reengineering was a fascinating tool enabling management organization is to overcome the crisis or sustain a strategic advantage and improve competitiveness. Considered the "organizational paradise": less loss of time (50 to 70% gain), cost savings of up to 30%, increased customer satisfaction and higher quality products and services. Hammer in [10] also refers to an organization without conflict and without social distinction, where motivation and happiness crisscross the climate prevailing between employees of the company. Today, reengineering is intended as a break with functional structures, described by Hammer and Champy in [9] Business "Taylorism" or bureaucratic or formalism and control procedures are the obstacles that make it impossible to adaptation to new environmental constraints. One of the goals of the business process reengineering is the time saved in processing customer requests or development of a product. In this time search logic, the job splitting units

between operators and is perceived as huge time loss generator (transfer time, the hierarchy of priorities, etc.).

2.2. Implementation of process reengineering

Among the important strategic decision a company can make is not to determine how to use information technology to set communications to improve its process, but which business processes, it must improve. It is recognized that each organization uses hundreds or even thousands of business processes [14], too, should determine which model improve to make the most of the technology or global information system. Indeed, when the company uses information technology to support the inappropriate process, it is more efficient to do things she should not do [15]. The consequence would be that the company becomes vulnerable to attacks from competitors who discovered the right business model [16]. Understanding and evaluating the performance of the organization's processes are essential before deciding to re-engineer. If the company seeks, deciding to re-engineer, reducing the time and saving costs related to the development of new products or services or delivery orders, it must first estimate the time and costs associated with these processes [16]. It is not wise to spend a lot of time and money to improve the processes that have little impact on its overall performance and turnover. It is recommended to determine which business processes are critical and deserve all the attention in the establishment of a new information technology, and then analyze how their improvement will support the implementation of the corporate strategy.

During the years 90, the computerization of a process was an optimization in itself. Today, the management by the process is no longer exclusively an approach of computer systems; the reorganization of labor significantly improves the performance of the process. However, the technologies of the information and communication maintain an important role since their contribution promotes the achievement of the process of highest performance levels [17].

The computer technologies such as EDI (Electronic Data Interchange), the RFID (Radio Frequency Identification), the smart sensors, etc. promote the automation of a processor of some of its activities. They are also used to organize the data from a process, which allows you to have access to the necessary information much more quickly. By having concrete data on the performance of the process, managers will be able to make the day-to-day decisions much more easily. In addition, they are released from a certain load of work, control of certain tasks that can be performed automatically by the system [18]. Another contribution of computer technologies is materialized by the systems of repositories of processes, which are particularly important in organizations, are structuring the process and who understand the modeling and documentation of the process architecture, facilitating the consultation and understanding that we can have the organization.

In addition, prior to rethink the process, a series of questioning should be conducted to wisely decide whether a re-engineering process or not recommended:

- Competition does significantly ahead of the company?
- Is the unstructured communication used in an abusive manner? (notes, email, etc.).
- Are the oppositions and conflicts present in the organization?
- Is there an extremely strong and unjustified frequency of meetings?

In what follows, we will study the processes of a logistic platform (PL) who wants to integrate new technologies to improve control of its activities in order to optimize the time, cost, ensure a better traceability of the flows, and reduce see eradicate the errors of salaries of its commands. The choice of performance indicators for steering the activities of the PL may be based on the KPIs proposed by SCOR.

III. PROCESS MODELING

3.1. Scope of SCOR & BPMN

The restructuring or reengineering of the process requires first a perfect knowledge of the whole of the process of the company, something that is hardly feasible in reality, or the need to use of modeling tools capable of describing faithfully the existing processes as well as their interactions with the process customers and/or suppliers. Among methods or modeling tools based on a process approach, we cite the SCOR model (Supply Chain Operation Reference Model) which has been put in place in 1996 by the Supply Chain Council (SCC), and BPMN (Business Process Management Notation).

The SCOR model has been developed to describe the business activities associated with all phases of satisfying a customer's demand. The model contains several sections and is organized around the six primary management processes of Plan, Source, Make, Deliver, Return and Enable (shown in Figure 1). By describing supply chains using thesis process building blocks, the model can be used to describe supply chains that are very simple yet very complex using a common set of definitions. As a result, disparate industries can be linked to describe the depth and breadth of virtually any supply chain. The model has been able to successfully describe and provide a basis for supply chain improvement for global projects as well as site-specific projects [19].

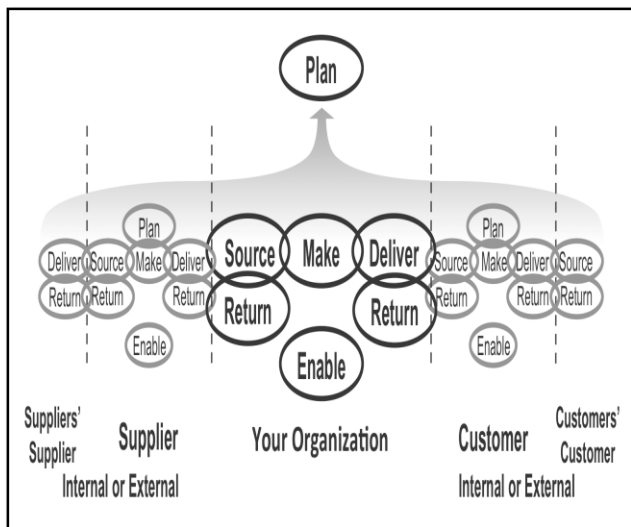


Fig.1. SCOR is organized around six major management processes [Supply Chain Council, 2012]

The BPMN (Business Process Management Notation) allows a representation standardized by the progress of the processes, allows the modeling of interactions and points of synchronization between processes.

3.2. Description of the PCD processes.

We distinguish, mainly three actors in this supply chain: suppliers, PCD, and customers. The mission of the suppliers is to truck the parts to the PCD. Considering these activities, the processes "Source" and "Deliver" of the SCOR model will represent the tasks of these suppliers. The PCD is the platform of cross docking. It has mission to answer at the request of her customers through the grouping and shipping the parts by boat. In the case of delay, the air transport can be also used but with a negative impact on the performance indicators of the PCD.

The modeled processes will be the process "source" which corresponds to the process of maritime containers reservation, the process "Make " which corresponds to the grouping and the reconditioning of the parcel to optimize the transported volume and the process " Deliver " which groups the preparation of commands, load, and shipping the parts, towards the customers.

3.3. SCOR & BPMN Model of the PCD

We model the processes of the studied supply chain, by basing itself on the levels 1, 2 and 3 of the SCOR model. This model allows giving a global view onto the inter-organizational functioning by describing all the processes, which interact between the platform, its customers, and its suppliers.

The level 3 of SCOR model is more complicated, because at this level, we have to model the interactions between the sub-processes.

To landing in the static aspect of SCOR model, we propose the utilization of the tool BPMN 2.0.

The initial modelling of macro process by using SCOR facilitated us the modelling, interactions between sub processes by BPMN.

The level 4 is not actively involved of the model SCOR. It is a question here of coming down at the level of the elementary activities which make up the processes of every entity of the SC.

The SCOR model does not propose here a toolbox or performance indicators. The reference [27], describes the modeling of the interactions between the PCD, its customers, and its suppliers, according to SCOR (Level 1, 2 & 3).

By basing on the preliminary modellings of the PCD by using SCOR. This one helped us to develop the BPMN model.

The reference [28], presents the modelling result by BPMN 2.0, the interactions between the macro and the micro process of the cross-docking platform (PCD), his customers and his suppliers.

IV. WEAKNESS OF THE PROCESS AND DISCUSSION OF RESULTS

The model presented highlights of the major weaknesses in the overall organization of the PCD between suppliers and customers. In effect, no visibility is possible on the operational process internal to the PCD, which creates a real difficulty in the management and synchronization of physical flows, between the suppliers and the plant. In fact, it has been demonstrated that the permanent flow of the goods is vital to the success of the cross docking and everything that can disrupt this flow will block the successful completion of the cross docking. The efficiency of the delivery of the goods is obtained by the coordination of the flow of information and of the physical flows. This is the importance of the visibility that allows you to succeed the implantation of the cross docking. In an environment of coordination such as the automotive industry, the sharing of information and, therefore, the visibility of process is a condition for the success of the continued replenishment and cross-docking [22-23].

V. IMPROVEMENT PROPOSITION: INTEGRATION OF THE RFID

5.1 Validation of the model by Simulation

The model developed by SCOR and BPMN showed us weaknesses in the management of business processes, notably on the synchronization of the physical and information flows.

To assure a better traceability and a good visibility on the business processes of the platform of cross docking, we propose the integration of the RFID.

This technology presents an opportunity to improve the operational performance of the PCD through a better management of the interactions inter-process.

The operational of the PCD can base themselves on the information communicated by the RFID tags to direct package to the good destination.

The reference [28] presents the PCD modelling after integration of the RFID, so in this model we find the various interactions and exchanges between the product and the other processes of the PCD.

In this paper, our models will be feigned by TIBCO business studio. Our objective is to feign the physical and informative flows, and to have comparisons between several scenarios to validate the proposed solution.

- Scenario 1: current State, without integration of the RFID.
- Scenario 2: simulation of the PCD processes, after integration of the RFID.

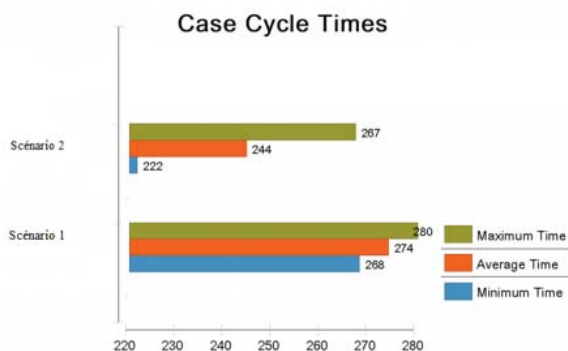
5.2 Results

We present the simulation result of the platform functioning before and after the integration of the RFID.

- Time of cycle

The figure 2 presents a comparison between the cycle times of the scenarios feigned in this article.

Fig.2. comparison of the orders treatment times before and after integration of the RFID in the PCD.



The comparison of the simulation cycle times, shows that cycle time, after integration of the RFID (Scenario 2), approached the optimum. Seen that, the time of the scenario 1 schematized in the graph is just of the platform operations. Then if we add the transport time, the difference between the first scenario and the proposal would be bigger.

- Costs

The figure 3, presents a comparison of the costs linked to scenario 1 and scenario 2.

The costs are expressed in EURO.

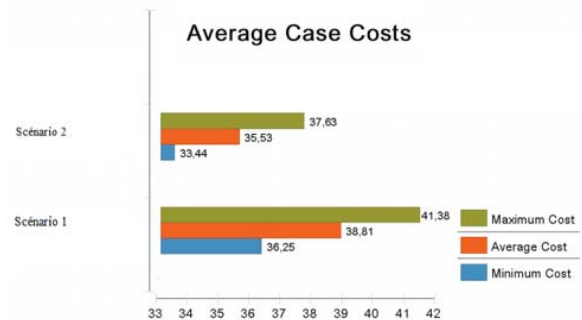


Fig.3. comparison of the platform costs before and after integration of the RFID.

The cost of the first scenario is really bigger than the value calculated by simulation, if we add the costs linked to the air repair. Of this fact we can conclude that by integration of the RFID we obtained a cost approached the optimum and thus we contributed to the improvement of the PCD performance.

VI. PROCESS APPROACH IN THE INTEGRATION OF THE RFID IN THE PCD: ORGANIZATIONAL PROCESSES REENGINEERING

Having demonstrated positive impact of the RFID in the improvement of the PCD operational performance. However the success of the integration of such technology can make a success only with an implementation of an organizational structure based on a revision of the existing processes and thus of a processes reengineering.

To measure permanently the efficiency of the set up system, we suggest creating a new process, under the name "pole of the performance control ". The input of this process will be the data recorded by the RFID tags, and the output of this process will be the measure of the KPIs beforehand identified by a selection of KPIs ranked by PCD leaders and level 3 of SCOR Model.

The existence of such process in the PCD mapping , will on one hand allow to restructure input and output of every process, and on the other hand to have a control of the process business of the PCD but also the relations between the suppliers and the customers .

CONCLUSION

In this paper, we examined the role of process re-engineering in the success of new technology integration projects. A real

case study of cross docking platform (PCD) for an automaker has been addressed by using the SCOR model and BPMN. The choice of this model is explained, firstly, by the fact that SCOR is one of the best tools to model the processes of PCD; secondly, SCOR proposes performance indicators (KPIs) that provide the piloting of the activity. While BPMN allows of landing the SCOR static aspect by an advanced management of the events and the return flows.

After modeling the PCD with SCOR & BPMN and identification of KPIs, we have detected some organizational weaknesses in the operational processes of PCD and processes between customers and suppliers.

We proposed the integration of the RFID in the PCD business processes management.

We validated the model developed by simulation.

We feigned the time and the orders processing cost in the platform, before and after the integration of the RFID.

The simulation results demonstrated the positive impact of the RFID over the deadlines and the orders of treatment time.

The fact of introducing the new technologies, in the SC, as the RFID, is not sufficient to reach the desired performance level.

We recommended a processes reengineering, through the addition of a new process in the PCD mapping. We proposed the name “pole of the performance control”. This process has role to guide the platform on the weaknesses of business process, through a permanent measure of the KPIs.

We also proposed the strategy of these KPIs choice.

We chose the KPIs of the PCD leaders and we compared them with the KPIS of the SCOR level 3. This study allowed having an exhaustive dashboard.

Our research perspectives are trying to continue to support the platforms in its process reengineering approach for integrating new technologies into its business processes.

We wish to generalize this approach on the SC set actors, to maximize the return on investments in the new technologies projects adoption.

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